A13485W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6: CONDENSED-MATTER PHYSICS

TRINITY TERM 2024

Thursday, 6 June, 2.30 pm – 4.30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Elemental sodium crystallizes in a body-centered cubic (bcc) structure at ambient pressure with lattice parameter $a_0 = 4.27$ Å. It undergoes a first-order structural phase transition to a face-centered-cubic (fcc) structure at a critical pressure $p_c = 65$ GPa. At a pressure just below p_c the cubic bcc lattice parameter is a=2.90 Å, and at a pressure just above p_c the cubic fcc lattice parameter is a'=3.65 Å.

(i) Sketch the three-dimensional conventional unit cells of the two crystal structures, just below and just above p_c, and (ii) determine the number of atoms per cubic unit cell in each case. (iii) Draw the (111) plane above and below p_c, indicating the positions of sodium atoms, and (iv) calculate the nearest-neighbour distance in this plane. [6]

(b) An X-ray diffraction pattern of a powder sample of sodium is collected using X-rays of wavelength λ=0.43 Å.

(i) State the appropriate diffraction selection rules and determine the positions of all observed reflections for a total scattering angle $0 < 2\theta < 25°$ at a pressure just below and just above p_c. (ii) Sketch the two diffraction patterns (intensity as a function of scattering angle 2θ) and label all diffraction peaks with their Miller indices, assuming that the atomic scattering factor is independent of the scattering wave vector and disregarding any additional instrumental effects. [9]

(iii) Starting from a pressure just above p_c and increasing the pressure to p_1=100 GPa, it is found that the third allowed diffraction peak as a function of increasing 2θ shifts by $\delta(2\theta) = 1.20°$. Find the value of the fcc cubic lattice parameter, a'_1, at p_1 to a precision of 10^{-2} Å. [3]

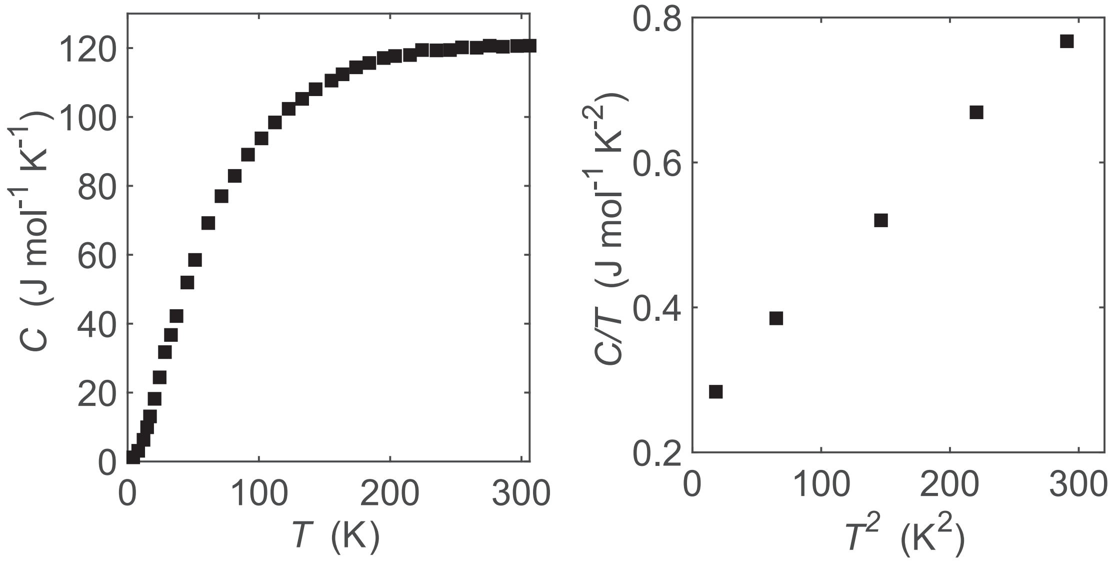
(c) Treating sodium atoms as hard spheres, determine the radius of a sodium atom in the two structures, just above and just below p_c. Hence, or otherwise, determine the change in density at the structural transition at p_c. [3]

(d) The compressibility is defined as $\beta = -\frac{1}{V}\frac{\partial V}{\partial p}$, where V is the volume and p is the pressure. Determine the compressibility of the bcc phase, assuming that it takes a constant value between ambient pressure to just below p_c. [2]

(e) Determine the mass of sodium which can be pressurized in an actual experiment when a powder sample is placed inside the cylindrical volume of a pressure cell with an inner diameter of 100 μm and a height of 1 mm (When the powder fully fills this cylindrical container the space between powder grains accounts for 35% of the total volume). [2]

[The molar mass of sodium is 23.]

2. (a) The temperature dependence of the molar heat capacity C of the three-dimensional metallic material $YbMg_2Bi_2$ is shown in the left panel of the figure below.



Explain the physical reason why the heat capacity saturates at high temperatures and relate the limiting value to the number of atoms per formula unit. [3]

(b) At low temperatures, the molar heat capacity C has the form

$$C = zN_{\mathrm{A}}k_{\mathrm{B}}\left[\frac{\pi^2}{2}\frac{T}{T_{\mathrm{F}}} + \frac{12\pi^4}{5}\frac{T^3}{T_{\mathrm{D}}^3}\right],$$

where N_{A} is Avogadro's number, z is the number of atoms per formula unit, T_{F} is the Fermi temperature and T_{D} is the Debye temperature. (i) Justify the functional form of the first term and derive fully the expression of the second term. [8]

(ii) Using the data in the right panel of the figure above, which is a plot of C/T versus T^2, deduce values for the temperatures T_{F} and T_{D}. (iii) Estimate the temperature T^* at which the two contributions to the heat capacity are comparable. [3]

(c) The same material is now deposited on a substrate as a two-dimensional thin film. Assuming that atoms vibrate only in the two-dimensional plane of the film, and that the sound velocity for longitudinal and transverse modes are equal, (i) determine the density of states as a function of frequency for normal modes of vibrations. (ii) Determine the high-temperature limit of the molar heat capacity of this two-dimensional crystal lattice. [3]

(iii) Show that in the low-temperature limit, the molar heat capacity of the thin film has the form

$$C = AT + BT^{\delta},$$

and determine the numerical exponent δ. (iv) Justify that A is related to the electronic density of states at the Fermi level $g(E_{\mathrm{F}})$ and deduce how B is related to the Debye frequency ω_{D}. [8]

[The relevant molar masses are Yb=173 g, Mg=24 g and Bi=209 g.

$$\int_0^\infty dx \frac{x^3}{e^x - 1} = \frac{\pi^4}{15}, \qquad \int_0^\infty dx \frac{x^2}{e^x - 1} = 2\zeta(3) \sim 2.4]$$

3. (a) The band dispersion of a material with a two-dimensional square crystal lattice is given by

$$\epsilon(k_x, k_y) = \epsilon_0 + 2t\cos(k_x a) + 2t\cos(k_y a),$$

where $t > 0$ is the isotropic hopping integral, a is the lattice parameter and x and y are along the square lattice edges.

(i) Sketch the band dispersion versus momentum along the path $(k_x, k_y) = (0, 0) \rightarrow (\pi/a, 0) \rightarrow (\pi/a, \pi/a)$. (ii) Determine the bandwidth of this band. [4]

(iii) Describe the shape of the constant energy contours in the two-dimensional momentum space inside the first Brillouin zone when the Fermi level is just below the band maximum, just above the band minimum, and exactly in the middle of the band. (iv) Determine the effective mass of the charge carriers when the Fermi level is just below the band maximum, and just above the band minimum, and comment on the character of the charge carriers in both cases. [8]

(b) (i) Show that when the Fermi level is just above the band minimum, the Fermi energy, ϵ_F, is related to the number of mobile electrons per unit area N/A as

$$\epsilon_{\mathrm{F}} = \epsilon_0' + \alpha \frac{N}{A}.$$

(ii) Determine ϵ_0', α in terms of ϵ_{F}, t, a. (iii) Deduce that the density of states near the band minimum is $g(\epsilon) = \beta A$ and determine β in terms of t, a, and fundamental constants. [9]

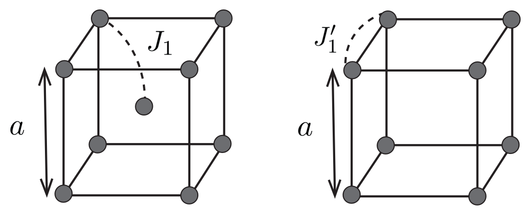
(c) (i) Using the above expression for ϵ_F, determine the carrier density when the Fermi level is just above the band minimum and the area enclosed by the Fermi surface is 1/100 of the Brillouin zone area, for the lattice parameter $a = 3.5$ Å. (ii) In this case, calculate the Hall voltage that develops across the device along the y direction, in the presence of a magnetic flux density of 1 T applied normal to the plane of the material and an electrical current density on the surface of 10 μA per meter flowing along the x direction. [4]

4. (a) Using Hund's rules (i) determine the quantum numbers L, S, J, and Landé factor g_J of the free ions Mn^{2+} ($3d^5$) and Tb^{3+} ($4f^8$). (ii) In both cases determine the magnitude of the magnetic moment (in units of μ_B) in an applied magnetic field at zero temperature. The Landé factor g_J is

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}.$$

[6]

(b) Magnetic moments with spin quantum number $S = 1/2$ and Landé factor g are arranged on the vertices of a body-centred cubic lattice of edge a, as shown in the left panel of the figure below.



The spin Hamiltonian is

$$\mathcal{H} = -\sum_{\langle i,j\rangle} J_1 \mathbf{S}_i \cdot \mathbf{S}_j + \sum_i g\mu_B \mathbf{B} \cdot \mathbf{S}_i,$$

where the first sum extends over all nearest-neighbour pairs $\langle i, j\rangle$ counted once, $J_1 > 0$ is the exchange constant, and the second term is the Zeeman energy in an applied magnetic field of flux density $\mathbf{B}$. In zero applied, field the exchange interactions stabilize ferromagnetic order below a critical temperature T_C.

(i) Show that in the *mean-field approximation*, each spin experiences an effective magnetic field

$$B_{\mathrm{mf}} = \lambda M + B,$$

where M is the magnetization (magnetic moment per unit volume) and determine λ in terms of J_1, g and fundamental constants.

(ii) Show that the magnetization at temperature T satisfies

$$M = M_0 \tanh\left(\frac{g\mu_{\mathrm{B}} B_{\mathrm{mf}}}{2k_{\mathrm{B}}T}\right),$$

and determine M_0 in terms of g and a. [6]

(iii) In the limit of small B and at temperature $T > T_{\mathrm{C}}$ in the paramagnetic phase, solve for M self-consistently, and deduce that the magnetic susceptibility is of the form $\chi = \frac{C}{T-\Theta}$. Determine Θ in terms of J_1, and C in terms of a and g. (iv) Relate Θ to the critical temperature T_{C}. (v) Determine J_1 for $T_{\mathrm{C}} = 30$ K. [8]

(c) A second material containing $S = 1/2$ ions arranged in a simple cubic lattice, shown in the right panel of the figure above, has a nearest-neighbour exchange interaction J_1' and orders ferromagnetically below $T_{\mathrm{C}}' = 20$ K. (i) Determine the value of J_1'. (ii) Sketch on the same graph the temperature dependence of the inverse susceptibility $\chi^{-1}(T)$ for both materials in the temperature range above their respective ordering transition temperatures. The cubic lattice parameter in both materials is the same. [5]